

off the laptop. The laptop wouldn't even reboot properly; I'd have to shut it down and then start it again. Another (proprietary) operating system I tried had fatal problems, which were only resolved by using the very latest version.

- The sound card is based on the Conexant CX20583 (Pebble HSF) chip-set, and uses the Intel HDA driver. ALSA does not support this device properly. Ubuntu 8.10 came with the Linuxant ALSA drivers, which, unfortunately, don't work on 10.04. So there's a faulty jack-sensing 'feature': when you plug in headphones, either the laptop speakers don't go mute and you hear sound from both, or your headphones never work at all. Other problems that you're likely to face:
- The touch-pad and/or the keyboard do not work. This is a very common complaint.
- Wi-Fi does not work. This is an issue with laptops that have a Broadcom Wi-Fi adapter, including MacBooks.
- If you have a PCMCIA card or slot, during hardware detection it may crash your system. This is not much of an issue on newer laptops.

The common solutions

For my first problem (the ACPI issue), I tried a lot of things which were fruitless. Then I thought, "What the heck, let's just get a new kernel!" Lucid, as of now, uses the 2.6.32 kernel. I downloaded the 2.6.34-RC4 kernel, and built Ubuntu packages from it. It's not that difficult; the steps I followed were (all terminal commands) as follows:

```
wget -c http://www.kernel.org/pub/linux/kernel/v2.6/testing/linux-2.6.34-rc4.tar.bz2
tar -xvf linux-2.6.34-rc4.tar.bz2
cd linux-2.6.34-rc4.tar.bz2
sudo apt-get build-dep linux-image
sudo apt-get install kernel-package fakeroot libncurses5-dev
cp /boot/config-`uname -r` ./config
make oldconfig
```

At this stage, you need to answer a few questions -- about which newly introduced features should be compiled. It's a good idea to build everything as modules (press 'm'), and for those which cannot be built as modules, just enable them. Next, run:

```
make-kpkg clean
fakeroot make-kpkg --initrd --append-to-version=custom kernel_image
kernel_headers
```

When this completes, you'll find two shiny new Debian packages in the upper level directory. Install them by typing the following command:

```
cd ..
sudo dpkg -i *.deb
```

The first time I booted with the new kernel, there was

kernel panic, with a message saying it couldn't find a root device. I found out that InitRamFS had not been generated. To generate that, type in the following:

```
sudo update-initramfs -c -k 2.6.34-rc4-custom
```

Replace 2.6.34-rc4-custom with your kernel version (the one you built). That's it.

While installing the new kernel solved the ACPI problem, it stopped the sound completely. Before the kernel update I was getting sound from both my speakers and my headphones, at the same time. With the upgrade, however, the headphones stopped working altogether.

After countless hours of searching, tinkering with *modprobe* files, manually loading drivers and more, with no results, I tried a Bluetooth headset, and it worked. So while my sound card may be non-functional, I have other ways to get sound. The perfect solution, it seems, is USB Audio.

USB Audio is basically a USB class. Therefore, it is completely plug-and-play, and does not require any drivers. What happens is that when you plug in a USB sound card, the computer sends a digital signal to the USB device. If the USB device is a sound card, it just converts the digital signal into an analogue one, and sends it over conventional mini-plugs. However, in the case of USB sound devices like USB speakers and headphones, the signal remains digital all the way to the actual speakers. Thus, you get digital sound, with no noise or distortion—a bit-by-bit perfect reproduction of the sound. All Bose speakers are USB, by the way.

USB sound devices aren't very expensive. You get generic sound cards on eBay for Rs 200 or less, and they sound just perfect. More expensive studio-quality sound cards like the Creative X-Fi USB come for something like Rs 4000-5000, while USB headphones and speakers cost around Rs 1000.

Now, for the solutions about Broadcom Wi-Fi. In the recent past, Broadcom used to have no Linux drivers, and you'd be forced to use Windows drivers with NDISwrapper. However, Broadcom has recently made available proprietary drivers for Linux, something like nVidia's and ATI's closed drivers for their video cards. While the exact installation steps differ from one distro to the other, the steps for Ubuntu and Fedora are given below.

In Ubuntu:

1. Go to **System->Administration->Hardware Drivers**
2. Choose the **Broadcom STA Wireless Driver**
3. Hit **Activate**.

Sometimes the driver does not show up among the **Hardware Drivers** choices. In this case, get your PC on the Internet somehow (via a wired link), or add the install CD as a source (**System->Administration->Software Sources**), and then run the following:

```
sudo apt-get update
sudo apt-get --reinstall install bcmwl-kernel-source
```

Now reboot your machine, and get back to the **Hardware**